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The Portable Emissions Tracer

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Abstract

Additive manufacturing processes release volatile organic compounds (VOCs), aerosols, and ultrafine particles that can accumulate in poorly ventilated environments and contribute to long-term respiratory or cardiovascular diseases. Industrial systems often include filtration or monitoring equipment, but affordable, real-time solutions are rarely available for non-industrial users such as hobbyists, students, and small labs.

We propose a **portable handheld emissions tracer** that integrates multiple sensors for ultrafine particulate matter (down to $1.0\text{ }\mu\text{m}$), formaldehyde and total VOCs, into a compact, battery-powered platform. The device provides up to four hours of operation, can function as either a handheld or stationary monitor, and enables remote access to air-quality data through a web interface.

Our research involved literature review, analysis of OSHA standards, and consultation with a campus 3D-printing lab. Based on these findings, we developed a sensor architecture that delivers accurate, real-time monitoring of hazardous emissions. By providing accessible alerts and data visualization, the system empowers non-industrial users to make informed decisions, reduce exposure risks, and improve additive-manufacturing safety.

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Introduction

As 3D printing technology becomes more affordable and accessible, concerns about its impact on operator health have grown. Both Fused Deposition Modeling (FDM) and Stereolithography (SLA) resin printers emit volatile organic compounds (VOCs), aerosols, and particulate matter during printing and post-processing. Research conducted by Chemical Insight indicates that prolonged exposure to these emissions can contribute to respiratory issues, skin irritation, and other health concerns [1]. However, because consumer 3D printing has only become widespread in the past decade, many personal and academic users remain unaware of these risks and often lack proper ventilation or emission monitoring.

Existing mitigation methods primarily rely on ventilating the area or using HEPA and carbon filters to capture hazardous particles and gases. While effective, there are limited ways for users to monitor their 3D printing environment. Users are in need of a generic air quality monitoring system. Although general-purpose air quality monitors are commercially available, most are designed for ambient indoor environments rather than the specific emission profiles of 3D printing. They often lack the sensitivity, calibration, or sensor range needed to accurately detect VOCs, specific gas such as formaldehyde and ultrafine particles (UFP) produced by FDM and SLA printers. To address these limitations, our project proposes a portable emission tracer—a compact, sensor-based monitoring system tailored specifically for 3D printing environments to detect and alert users of hazardous emission levels. The device integrates multiple sensors on an ARM embedded system (Raspberry Pi Zero 2) using an IoT framework to collect real-time VOCs emission index, formaldehyde concentration and number of particles of particulate matter (PM). The system will act as a web server to transmit the data to a client upon request for real-time monitoring. This system provides an affordable, user-friendly solution that promotes safe operation in academic labs, maker spaces, and home workshops while complying with OSHA and ANSI safety standards. The current scope of the air monitor can adequately collect data (concentration of VOC'S in the environment) and display the results on the screen.

Background Research

3D Printing, whilst an immensely powerful tool, does not come without drawbacks. The use of this technology can cause the buildup of harmful substances that when inhaled can lead to long term health problems as stated by the CDC [2]. Multiple studies have confirmed that desktop 3D printing processes generate both volatile organic compounds (VOCs) and particulate matter, especially ultrafine particles (UFPs) [3], [4], [6], [10], [12]. In an article by Azimi published in 2016, UFP and VOC emissions are quantified. It is suggested that users need to be cautious when using 3D printers in poorly ventilated spaces, or without a proper filtration system [4].

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Commercial use has been around for some time, and many who are already in industry are aware of the risks that are involved when working with diverse types of machines and the different hazards associated with them. The bigger issue though is that many who use 3D printers commercially are ignorant of the possible ramifications of continuous use.

The main approach to helping keep atmospheric conditions acceptable is through ventilation and proper air flow within a system. This does not guarantee that a work or living environment is safe to breathe in even with an active ventilation system present. To help monitor this, sensors for common chemical offputs like formaldehyde were developed as well as sensors for particulate matter and ultrafine particles (UFP). Most of these sensors are used in settings such as industrial complexes and laboratories and are not easily accessible to ordinary households.

Tables and Figures

Since working with materials that emit hazardous gases is required, there are exposure limits according to safety standards from organizations such as OSHA, CDC, and NIOHS. For particulate matter, as stated by the CDC in their Particulate Matter Guidance, assessment of PM exposure to establish an acceptable range is complex; they based it on WHO and the results for PM exposure limit are in Table 2 (CDC PM). In Table 1, the common VOCs that are produced during 3D printing processes, and Table 4 highlight the most toxic gases and hardest for humans to detect because they are odorless. The two graphs below show how much VOCs and particulate matter, size and concentration are emitted during the printing process. For Figure 2, we can see that at the beginning of the printing process, most of the particles are between the size of 7 to 50 nm, but these smaller particles will combine to form larger particles leading to a significant drop of the amount of the smaller size particles. Figure 1 shows the results directly from the health hazard evaluation in a specific manufacturing facility [6].

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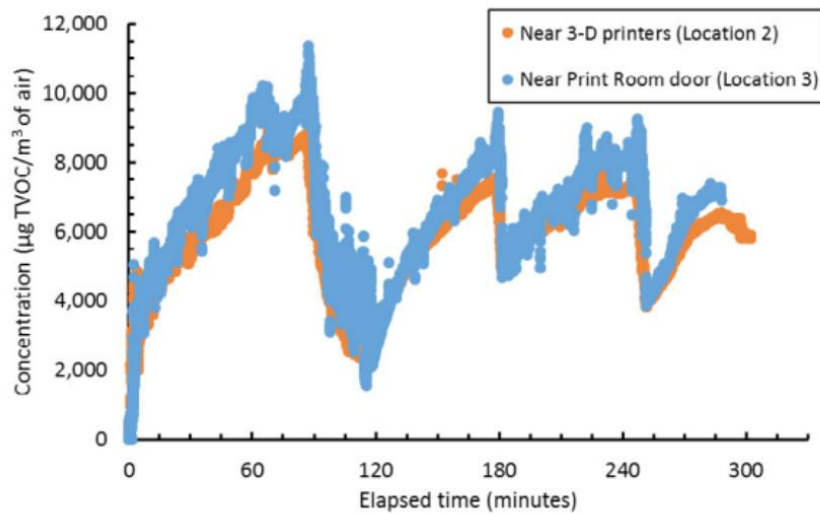


Figure 1: Reprinted from Volatile Organic Compounds during printing processes [6]

Location 2 is next to a 3D printer, and Location 3 is next to a doorway.

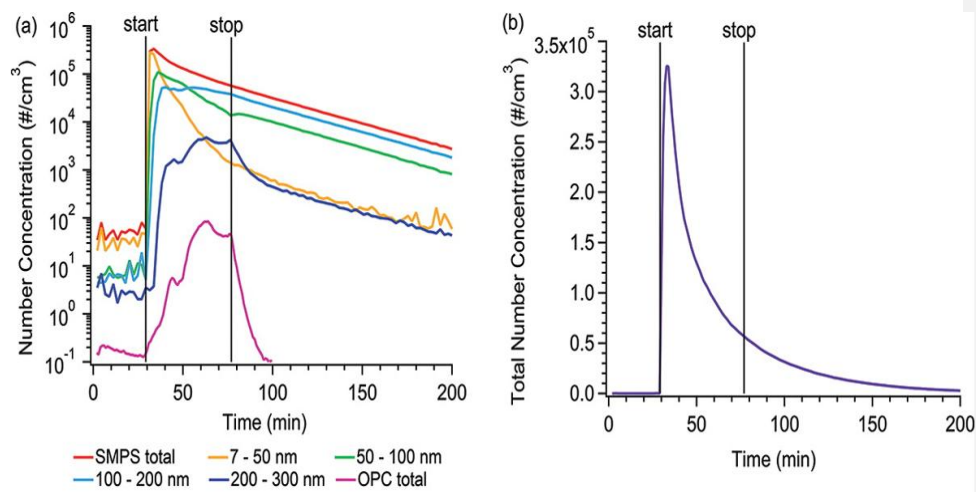


Figure 2: Particle size distribution in the Print Room during the day while printing only with PLA filaments. The average particle size was about 44 nm. [6]

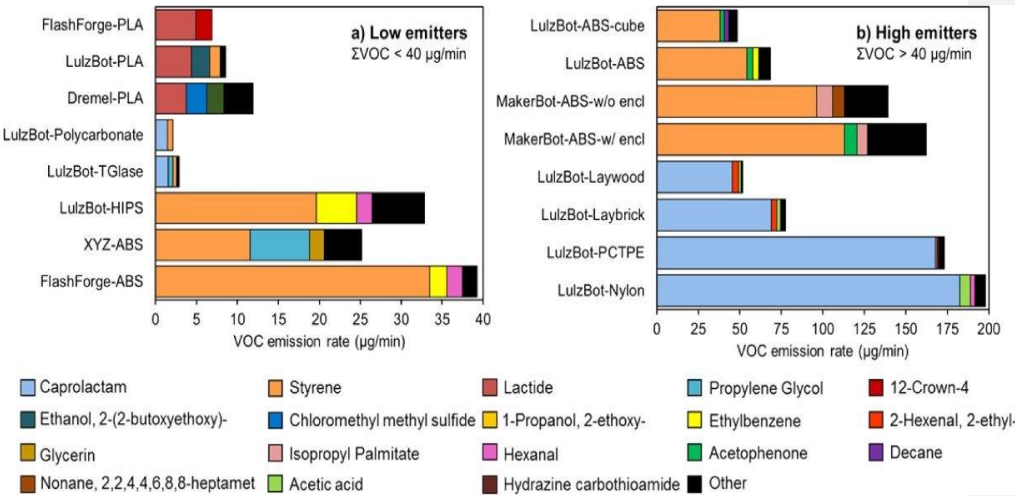


Figure 3: Summary of estimates of individual speciated VOC and total VOC emission rates from each of the 16 printer and filament combinations. [4]

Table 1: 3D Printing VOCs Overview

Compound	Sources	Health Impacts	Common Uses
Acetic Acid	Resin & FDM	Irritant	Vinegar, chemical reagent
Acetone	Resin & FDM	Impacts nervous system	Nail polish remover, PVC, solvent
Alpha-pinene	Resin & FDM	Generally safe	Fragrances, flavorings
Caprolactam	FDM	Sensitizer	Artificial textiles, nylon, resins
Cyclopentanone	Resin	Irritant	Fragrances, resins, adhesives
Dioxane	FDM	Reasonably anticipated to be a carcinogen	Inks, adhesives, solvent
Ethanol	Resin & FDM	Impacts nervous system	Solvent, fuel, liquor
Ethyl Acetate	FDM	Irritant	Nail polish remover, paints, adhesives
Ethyl Hexanol	Resin & FDM	Irritant	Plasticizer, fragrances, paints, resins

Ethyl Methacrylate	Resin	Sensitizer	Plexiglass, acrylic nails, dentures
Formaldehyde	Resin & FDM	Known to be a human carcinogen	Embalming, adhesives, resins
Hydroxypropyl Methacrylate	Resin & FDM	Sensitizer	Dental fillings, adhesives, resins
Isobornyl Acrylate	Resin & FDM	Sensitizer	Photoinitiator, inks, adhesives, resins
Isopropyl Alcohol	Resin & FDM	Respiratory irritant	Disinfectant, solvent
Lactide	PLA	Generally safe	PLA, drug delivery, packaging
Limonene	FDM	Generally safe	Fragrances, cleaners, solvent
Methyl Ethyl Ketone (MEK)	Resin & FDM	Impacts nervous system	Inks, paints, adhesives, resins, rubbers
Methyl Methacrylate	Resin & PLA	Sensitizer	Resins
Nonanaldehyde	Resin	Irritant	Fragrances, flavorings, polishes
Propylene Glycol	Resin & FDM	Generally safe	Antifreeze, food preservative, cosmetics
Styrene	Resin & ABS, HIPS	Reasonably anticipated to be a carcinogen	Styrofoam, latex, insulation, resins
Tetrachloroethylene	PLA, ABS, PC	Reasonably anticipated to be a carcinogen	Dry cleaning, degreaser, solvent
Toluene	Resin & FDM	Impacts nervous system	Gasoline additive, urethane, TNT
Xylene	Resin & FDM	Impacts nervous system	Concrete sealer, lubricant, solvent

Table 2: Minimum Risk Levels for Hazardous Substances

Formaldehyde			
Inhalation	Acute (1-14 days)	0.04 ppm	0.049 mg/m ³
Inhalation	Intermediate (4-364 days)	0.03 ppm	0.03684 mg/m ³
Inhalation	Chronic (365 days and longer)	0.008 ppm	0.00982 mg/m ³
Propylene Glycol			
Inhalation	Intermediate (4-364 days)	0.009 ppm	
Xylene, Mixed			
Inhalation	Acute (1-14 days)	2 ppm	

Inhalation	Intermediate (4-364 days)	0.6 ppm	
Inhalation	Chronic (365 days and longer)	0.05 ppm	

Table 3: PM Screening Values: World Health Organization Particulate Matter Air Quality Guidelines (AQG)

PM Air Pollutant Metric	WHO
PM10	45 µg/m3 (24-hour) † 15 µg/m3 (annual)
PM2.5	15 µg/m3 (24-hour) † 5 µg/m3 (annual)

† These screening levels reflects the numeric value of the WHO AQGs for 24-hours
 PM2.5: Particulate matter with diameter less than 2.5 micron
 PM10: Particulate matter with diameter less than 10 micron

Table 4: Most Notable Volatile Organic Compounds during Printing Process

VOCs	Typical filament / Process source	Health / Toxicity notes	Relative abundance / Behavior
Styrene	ABS, HIPS (high temperature)	Intensified by high temperature Irritant to the respiratory system	Frequently found in ABS, the highest VOCs
Caprolactam	Nylon (PA) filaments	Breakdown by high temperature Irritant to the respiratory system	Detected in Nylon, the major VOC in Nylon
Formaldehyde	Detected from ABS, PLA, nylon and some resins	Known human carcinogen; Irritant to the respiratory system	Frequently detected in ABS filament and during some post processing of
Acetaldehyde	ABS, PLA, nylon, some resins	carcinogenic by IARC Group 2B Irritant to the respiratory system	often only detected in post processing studies
Benzene	Trace in some ABS, HIPS and resins	Known human carcinogen Irritant to the respiratory system	usually low concentration, but in response to high temperature process
Toluene / Ethyl Benzene / Xylene (BTEX)	ABS, some resins, certain composite filaments	CNS effects, irritation to the respiratory system	Detected and abundant in printing and filament

injection molding	PLA filament (thermally degraded stress cracks, brittle)	Resistant to chemicals	Comparison of PLA (active) vs PET (inert)
Methyl Methacrylate	Resins, PLA processing (PLA/PLA (plastic))	Irritant, fumes/respiratory; low impact	Good during PLA processing (high temp)
Benzaldehyde derivatives	Various filaments and resins	Serious irritation	Fishy smell, chemical smell, and odor
Other aldehydes	Multiple filaments/resins	Irritant, a high VOC	Common in many products, often co-

When the product manufacture for the air monitor systems, the following products were selected to representation the common features available list the systems, describe table 1

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Table 5: Comparison to two existing Products

Device/ Aspect	Description	Limitations	Key Insight for Our Design
Existing Solutions	monitors are bulky, expensive, non-IOT	expensive, non-IOT	Need for a small, IoT-
AirGradient ONE	Open source, IoT, and	Assembled, requires	Imp IoT bus or app source
Temtop	Low cost, handheld, and portable	Lacks IoT integration	More affordable, rugged
Our Approach	Compact, IoT, and portable	Using Raspberry Pi zero	Added IoT capabilities

Project Management

Gantt chart here

	Assignments	Signature
Forrest Austin		
Leanna Wildes		

Wilfred Talle		
Anh Phung		
Bota Yvan Brilliant		

Technical Information

Used to determine specifications |

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Table 6: Design Objectives

Customer Requirements	Description	Priority
Data Logging and Analytics	Record data and analyze air quality	1
Following of Regulations	The concentration limit must adhere to current industry standard	1
Life Cycle	How long will this run with acceptable accuracy	4
Portability	Compact and can fit seamlessly into any 3D printing set up	2
Alert Mechanism	Trigger warnings when emission concentration reaches the limit	3
Cost	Affordable price for the general publics and education users, being able to compete with current product	2
Real-Time Monitoring	Continuous monitor of air quality throughout the printing process	1
Detection of Emission	Identify VOCs, ultrafine particles (PM1.0, PM2.5, PM10), Formaldehyde	1
Accuracy	Accurately detect the concentration so the alarm can go off and let user know	1

Note: For priority, 1 is the most important, and 4 is the least important.

Words about the prototype | schematics

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The Air monitor system mostly deals with airflow monitoring inside an electronic insulated enclosure. Therefore, computer fluid dynamics as well as thermodynamics principles will be adequate for the report of precise calculations of the following parameters influencing the Airflow of a system:

The Volume flow rate: defined in CFM represents the volume of air per minute an external device can produce per minute in an environment. This parameter will be used to define the restrictions and the necessary RPM needed for the fan choice.

Dynamic Pressure: it represents the kinetic energy per volume gained by the airflow. It will be useful to identify the pressure the airflow would gain before being brought to zero velocity without any losses.

Flow Velocity: It's necessary to determine the air inlet flow velocity to meet the velocity of airflow requirements of the sensors SEN 55 which needs a maximum airflow velocity less than 1m/s.

ASSUMPTION:

Considering the scale and control volume of the enclosure, although the system is not adiabatic, we consider the volume to be constant supposing that the fluid velocity to be subsonic which favors a slight change in density of the fluid.

We consider the system to be pressured, and temperature dependent due to energy loss and dissipation in the form of heat by the battery, wires, raspberry pi, and sensors. Considering that variation, we consider the system to be in a transient state, therefore the mass flow rate will depend on the density of the fluid and will change over time. Furthermore, we will consider the initial conditions of pressure and temperature to be under the conventional environment state.

CONSTRAINTS AND PARAMETERS

Several constraints were determined by the manufacturers using the following estimation

Total heat: $Q = \text{dissipated power from raspberry pi} + \text{Battery}$

Total heat: $Q = 16 \text{ W}$

Critical temperature reach by the Raspberry pi: $T = 85 \text{ C}$

Battery critical temperature: VARIABLE

Recommended Airflow condition for SEN 55: Velocity < 1ms

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Design Calculations for SEN55

Considering the main goal of the Emission tracer, the suction of air and the cooling performance will be affected by the fan properties such as the RPM, the volume flow rate, and the flow velocity.

Applying fluid mechanics properties for compressible fluid, we need to determine the nature of the flow velocity using the Mach number being the ratio between the flow velocity and the flow of sound in the fluid:

$$M = \frac{V}{a}$$

M: Mach number

V: Flow velocity (m/s)

a: Velocity of sound in fluid (m/s)

$$a = \sqrt{\left(\gamma R \frac{T}{M}\right)}$$

γ : specific heat ratio (equals 1.4 for dry air at room temperature)

R: ideal gas constant (J/kg·K)

T: absolute temperature of fluid in Kelvin (K)

M: molecular mass of fluid (kg/mol).

Results

Considering our system being a fixed control volume:

$$\gamma = 1.4$$

$$R = 8.314 \text{ KJ/Kmol.K}$$

$$T = 20 + 273 = 293\text{K}$$

$$M = 28.96 \text{ Kg/Kmol}$$

$$a = \sqrt{\left(1.4 \cdot 8.314 \cdot \frac{293}{28.96}\right)}$$

$$a = 10.85 \text{ m/s}$$

Therefore, with the critical velocity being 1 m/s

$$M = \frac{v}{a}$$

$M \approx 0.1 < 0.3$ THEN the fluid can be treated as a subsonic incompressible fluid, but due to the change in temperature and pressure, the change in density cannot be neglected.

The general equation given for the mass flow rate is:

$$m = \rho \times V \times A$$

Air being an ideal gas give us the right to determine the density with the following expression:

$$\begin{aligned}\rho &= \frac{P}{RT} \\ \Rightarrow \frac{dm}{dt} &= \left(\frac{P}{R \cdot T} \right) \times V \times A \\ \Rightarrow V &= \left(\frac{\left(\frac{dm}{dt} \right) \cdot R \cdot T}{(P \cdot A)} \right)\end{aligned}$$

Average Dynamic Pressure:

$$Dp = \frac{1}{2} \rho v^2 = 7.6 \text{ pa}$$

Outlet velocity (Determined using Energy conservation equation)

$$v_{out}(t) = \frac{\dot{m}_{in}(t) + V \frac{\rho(t)}{T(t)} \frac{dT}{dt}}{\rho_{out}(t) A_{out}}.$$

From the simulation results the outlet velocity converges towards 1.4m/s

$$\dot{m}_{out}(t) = \dot{m}_{in}(t) + V \frac{\rho(t)}{T(t)} \frac{dT}{dt}.$$

Exposure Concentration:

$$C = \frac{ER}{AV_m}(IN_m)$$

C ($\mu\text{g}/\text{m}^3$) -estimated exposure concentration;

ER ($\mu\text{g}/\text{h}$) -emission rate determined from the chamber study;

V_m (m^3) -modeled volume;

A : -number of 3D printers in the modeled volume

N_m (h^{-1}) – the air change rate of the modeled volume.

Continuity equation:

Considering the conservation of mass flow rate for compressible fluid we have

$$\partial \rho / \partial t + \nabla \cdot (\rho \mathbf{v}) = 0$$

Considering our system these are the parameters for our initial and final conditions:

Initial Conditions: $T = 20^\circ \text{C}$ $P = 101.37 \text{ Kpa}$

Final condition: $T = 80^\circ \text{C}$ $P = ?$

Due to the fixed state of the enclosure and the absence of external factors (piston), we have an isochoric System (constant volume) therefore using ideal gas properties:

$$\begin{aligned} \frac{P_1}{T_1} &= \frac{P_2}{T_2} \Rightarrow P_2 = \frac{P_1 T_2}{T_1} \\ &\Rightarrow P_2 = \frac{(101325 \cdot (80 + 273))}{(20 + 273)} \\ &\Rightarrow P_2 = 122074 \text{ pa} \end{aligned}$$

Using the continuity equation: $\dot{m}_1 - \dot{m}_2 = \frac{dm_3}{dt}$

$$\rho_1 v_1 A_1 = \rho_2 v_2 A_2 + \rho_3 A_3 V(\text{enclosure})$$

$$\frac{P_1 v_1}{R T_1} \cdot A_1 = \frac{P_2 v_2}{R T_2} A_2 + \frac{P_3 v_3}{R T_2} A_3$$

Since $\rho = \frac{P}{RT}$

Due to our sole concern related to the flow velocity released from the fan to the enclosure, we will consider the inlet and outlet of the fan inside for the continuity equation

$A_1 = A_2$ since the flow is going through a uniform fan area

$$\Rightarrow \frac{P_1}{T_1} \cdot v_1 = \frac{P_2}{T_2} \cdot v_2 \quad v_2 \leq 1 \text{ m/s} \text{ Knowing } v_1 \text{ we can produce an airflow to induce } v_2 \Rightarrow$$

$$v_1 = \frac{(P_2 \cdot T_1)}{P_1 \cdot T_2} \cdot v_2 = 0.99 \text{ m/s}$$

Therefore, with an initial flow velocity less than 1 m/s, the enclosure due to its large volume

The Area of flow for an axial fan will be:

D(ring) = 30 mm, D(hub) = 18.5 mm

$A = \pi \frac{D^2}{4}$ For circular cross section

A(ring) = 0.0007069 m²

A(hub) = 0.0002688 m²

A = A(ring) - A(hub)

A = 0.0004381 m²

Formaldehyde average concentration emitted from SLA printers

The total concentration emitted by 3D PRINTING will be necessary to give an estimation of the amount of Vocs exposed in the manufacturer plant and determine the battery requirements such as the power, current and voltage that will be used to optimize the availability of the emission tracer in full time.

For a 6-hour resin printing session with 15 minutes of cleaning and 30 minutes of curing, the Formaldehyde emitted over the session is:

$C = (6 \text{ hours} \times 0.006 \text{ mg/hr}) + (0.25 \text{ hours} \times 0.012 \text{ mg/hr.}) + (0.5 \text{ hours} \times 0.012 \text{ mg/hr.})$

C = 0.045 mg

The average US room size: 11 x 12 x 8 [ft] which is 1056 [ft³] or 321.87 [m³] so the concentration for Formaldehyde is 0.0015 mg/ m³ with no air change rate per hour. For the success of sensor performance in sensing emissions, we may use different parameters related to fluid mechanics to maximize the flow of emission into the cavity of the gas detectors. Then, to find the necessary dimensions of the inlet pressure, we may use the continuity equation.

With the (Q been the air flow, V the flow velocity, A the cross-sectional area of the path taken. For a drop test, the identification of axial stresses present on the housing should be considered with the formula.

Calculating Required CFM, Current

Considering the compressible nature of air, its density, pressure and temperature condition might impact the volume flow rate calculation. Considering the change in temperature estimated from 25 C to a critical temperature of 85 C, the flow velocity will be given by:

$$V = \frac{Q}{A}$$

Q - the air flow (m³/s)

V - the output flow velocity, (m/s)

A - the cross-sectional area of the flow (m²)

Commented [LW10]: Mention before calculations that the purpose of all of the calculations is to determine formaldehyde emitted

Commented [LW11]: extrapolate on specifically what these calculations are being done for, in this case the fan

Commented [WT12]: @Anh Phung the area is the one for the flow and it's going to be the area of the ring - area of the hub of the fan so the free space between the blade where the airflow is going through.

$$Q = V \times A$$

$$Q = 1 \cdot 0.0004381$$

$$Q = 0.0004381 \frac{m^3}{s}$$

$$Q = 0.0004381 \frac{m^3}{s} \times 2118.88 \frac{CFM}{\left(\frac{1}{s}m^3\right)} = \mathbf{0.92 \text{ CFM}}$$

Using a 30 x30 x8 mm axial curved fan, it can produce a minimum rotational speed of 5000 RPM with a recorded corresponding CFM of 0.84 by the manufacturer.

Using Fan Law 3, we can relate the power required to the CFM desired, the minimum CFM and power produced.

$$Hp2 = Hp1 \times \left(\frac{CFM2}{CFM1}\right)^3$$

From MEGA suppliers, the rated input power varies from 0.4 to 0.9 W with a CFM range between 0.84 to 1.39 CFM for 30 x 30 x7 mm fans. From the fan various applications and the enclosure size we believe that our input power and a CFM should be as low as possible to better control power consumption and the flow velocity.

Hp2: power required

Hp1: input power (minimum value used)

CFM2: desired Volume flow rate

CFM1: input volume flow rate (minimum CFM used)

$$Hp2 = 0.0005438486 \cdot \left(\frac{0.92}{0.84}\right)^3$$

$$Hp2 = 0.0007145 \text{ hp} \approx 0.0007145 \cdot 743 \text{ W} = 0.531 \text{ W}$$

$$I = \frac{P}{V} = \frac{0.531}{5} = 0.1062 \text{ A}$$

CONCENTRATION OF MEMS HCHO

Converting PPM to $\frac{mg}{m^3}$

$$\text{Concentration} = \frac{1}{24.45} (\text{molecular weight} \times \text{concentration in ppm})$$

The range of concentration of the MEMS HCHO sensor in $\frac{mg}{m^3}$

Commented [WT13R12]: we will measure the hub experimentally

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$$\text{Concentration in HCHO} = \frac{(M(H) \times 2 + M(C) + M(O)) \times 3}{24.45}$$

$$\text{Concentration in HCHO} = \frac{(1.0078 \times 2 + 12.011 + 15.999) \times 3}{24.45}$$

Molar masses: $M(H) = 1.0078 \text{ g/mol}$, $M(C) = 12.011 \text{ g/mol}$, $M(O) = 15.99 \text{ g/mol}$

Concentration HCHO = 1.23 mg/m³

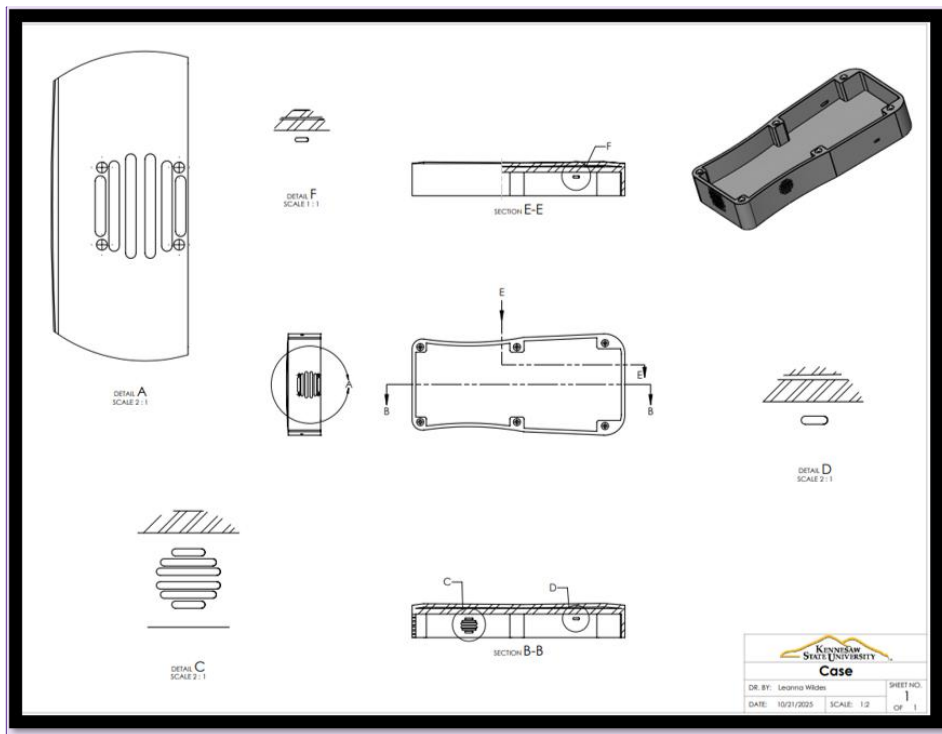


Figure 6: Part Drawing of Bottom of the Case

Power Supply & Battery Unit — Testing and Validation

Observation

The power supply subsystem consists of a **7.4 V / 10 000 mAh Li-ion pack**, a **TP5100 dual-cell charging module**, and **buck converters** for 5 V and 3.3 V regulation. The setup must safely

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charge the battery, deliver regulated voltages under dynamic load, and protect the system from overcurrent via a 3 A / 5 V fuse.

Battery Test Equipment

Equipment	Appearance	Description
Digital Multimeter (DMM)		Used to measure voltage, current, and resistance throughout the power circuit. It verifies supply levels, detects voltage drops across loads, and confirms proper continuity in wiring.
12v DC Adapter (2A rated)		Provides a stable input source to power the TP5100 charging module. It emulates the external power adapter that would be used during system charging and testing.
Oscilloscope		Measures and visualizes voltage ripple, transient response, and noise on the DC outputs. It ensures that the buck converters maintain smooth and stable output suitable for sensitive electronics like sensors and microcontrollers.
3 A Fuse		Provides overcurrent protection and safety isolation during testing. The fuse prevents damage to components in case of wiring faults or short circuits.
IR Thermometer		Monitors temperature rise on critical components such as the TP5100, buck converters, and battery pack during load testing. This ensures that the design stays within safe thermal limits.

Resistive Loads		Used to simulate the system's power consumption under different operating conditions. These loads help evaluate voltage regulation, thermal behavior, and runtime performance of the power supply.
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Battery Test Procedure

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1. **Initial Verification:** Check wiring continuity, polarity, and module orientation.
2. **Charging Test:** Apply 12 V input to TP5100 and observe LED indicators; measure charge voltage at $8.4 \text{ V} \pm 0.05 \text{ V}$ during CV stage.
3. **Battery Output Check:** Disconnect supply and record pack voltage between 6.5–8.4 V.
4. **Buck Converter Setup:** Calibrate 5 V converter output to 5.10 V and 3.3 V converter to 3.31 V under no load.
5. **Load Testing:** Incrementally increase current draw (0.5 A $\rightarrow$ 3 A) while monitoring voltage drops, ripples, and thermal rise.
6. **Protection Validation:** Simulate short circuit at output and verify 3 A fuse trip behavior and recovery.
7. **Charging While Operating:** Power the TP5100 and load simultaneously to confirm system-level stability during charge–discharge overlap.

Battery System Test Outcomes

- TP5100 required **12 V input** for effective 2S charging; achieved full 8.4 V output.
- Battery maintained **7.6 V nominal** under 15 W system load.
- 5 V rail stayed between **5.06–5.12 V**, 3.3 V rail between **3.28–3.33 V**.
- Ripple measured $< 40 \text{ mV}$ (5 V rail) and $< 25 \text{ mV}$ (3.3 V rail).
- The power system supported **Raspberry Pi + sensors for 1.9 h runtime** on full charge.
- Thermal rise on buck converter $< 24^\circ\text{C}$ above ambient.
- Fuse successfully isolated circuit overload, confirming overcurrent protection.

Battery Performance Benchmarks

- Input voltage: **12 V $\pm$ 5%**
- Charging voltage: **8.4 V $\pm$ 0.05 V**
- Regulated outputs: **5.10 $\pm$ 0.05 V, 3.31 $\pm$ 0.03 V**
- Ripple: **$\leq$ 40 mV p-p**
- System runtime: **$\approx$ 1.9 h @ 15 W**
- Thermal limit: **$< 45^{\circ}\text{C}$**
- Overcurrent trip: **3 A fuse activates < 2 s**

Battery test Conclusions

The power circuit demonstrated **stable operation, safe charging, and adequate regulation** for all modules. Voltage rails remained within tolerance, with minimal ripple and acceptable thermal performance. The design meets electrical reliability and safety expectations for portable operation.

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Sensors Accuracy and Precision

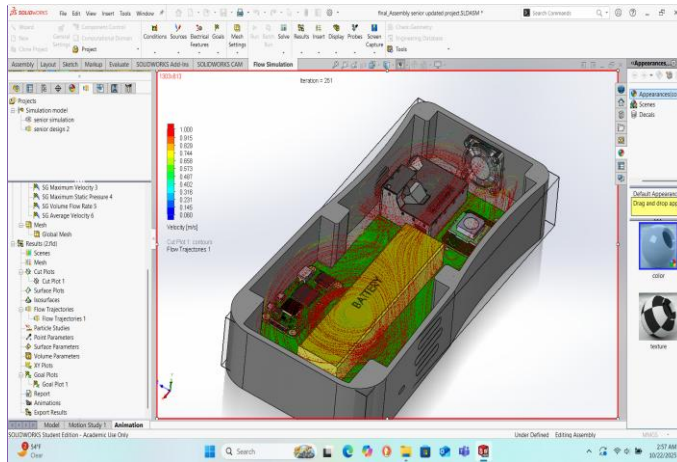
Computer Fluid Dynamic Simulation

The air monitor projects use the environment pressure to optimize the airflow in the enclosure using an axial fan with a velocity of 5000 RPM at maximum power. Due to the power dissipated by the battery, Raspberry pi Zero 2 and the sensors, the enclosure might face an increase of temperature over time equivalent to approximately 15W. Due to the change of pressure and temperature over time inside the enclosure, we consider the enclosure system to be in a transient state for the determination of the flow velocity and volume flow rate. The fan should produce an airflow to ensure the cooling process and the proper intake process of the environment flow into the system to favor the sensor chemical reaction to collect the different data and concentration of VOC'S present in the flow system.

Our goal over this flow analysis is to validate the choice of the fan used to optimize the airflow in the system to prevent overheating and favor the collection of accurate concentrations and data of VOC'S present in the enclosure for the air monitor system. Using SolidWorks Flow simulation, we might present the various results issued from several iterations of the software

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considering an external simulation with the environment pressure as our boundaries for both the inlet and outlet in the system to approach real conditions.



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Previous Results from this design configuration

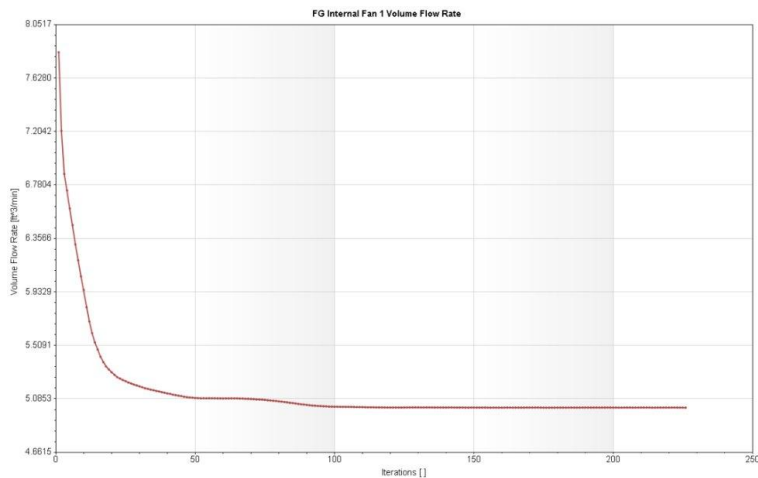
The plot indicates a flow velocity of 3m/s as the maximum value of flow produced by the fan at its optimum power along the axis of flow. Due to the static pressure that counters the airflow at the side, the velocity of the flow decreases between 0.915 and 1 m/s that fairly meet the sensors SEN55 requirement for the airflow.

Results from the optimized design configuration

From the requirement provided by the SEN 55 sensor manufacturer, a flow velocity less than 1 m/s is required for a stable and effective process of data collection. From our calculation, the input airflow velocity should be equal to the critical one given for the SEN 55 to ensure proper ventilation and give enough time to the sensors to record the VOC'S and formaldehyde concentration. Using the Reynolds number formula given by:

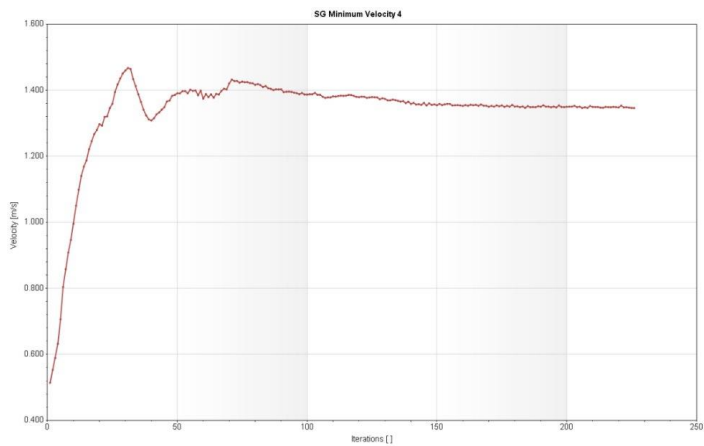
$$Re = \frac{\rho \cdot V \cdot L}{\mu} = \frac{0.987 \times 1 \times 0.203}{2.07 \times 10^{-5}} = 9.68 \times 10^3 > 4000 \text{ (Turbulent)}$$

Turbulent flow generally exhibits a parabolic distribution of their flow velocities which tends to be higher at the central axis of flow and lower at location distant from that axis. That property positively impacts and eases the placement of the sensors to meet the airflow requirements.



GRAPH 1: Fan volume flow rate convergence

This plot shows the convergence of the maximum volume flow rate produced by the internal 30X30X8 axial fan while being subjected to the environment pressure in the inlet. As the number of iterations increases, the value of the flow rate becomes constant and converges to approximately 5 CFM at maximum performance.



GRAPH 2: Minimum flow velocity produced by the fan at maximum performance

This displays a flow velocity value issued from the fan while subjected to a maximum static pressure (pressure preventing the airflow) in the enclosure. That value converges to a velocity less than 1.4 m/s which is still ideal for the airflow in SEN55 and formaldehyde sensors

Performance results:

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FINITE ELEMENT ANALYSIS

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Our goal, using the Finite Element Method (FEM), is to evaluate the impact of major external and repetitive loads on the enclosure material, geometry, and the fasteners used for the final assembly. Based on the portability of the enclosure, our team has identified several potential misuse scenarios and accidental load cases:

- **Drop from height:**
Because a fall introduces significant potential energy due to gravity, the resulting impact can generate compressive stresses that exceed the material's yield strength. This may cause critical damage to internal electronic components. The impact could also create medium to severe cracks in the enclosure, which may grow over time due to stress concentration.
- **Repetitive loads:**
Frequent handling of the enclosure such as continuous pulling, tightening, crushing, or separating of the top and bottom sections can lead to deformation of both the housing and the fasteners. Repeated tensile and compressive forces may cause long-term deformation of the enclosure cross-section and gradual damage or loosening of the screws.

Considering these potential misuse conditions, we will employ the following simulation methods:

❖ Static Simulation

This method will help determine the effects of compressive, shear, and tensile forces on the enclosure and fasteners. It will allow us to assess the suitability of the fasteners under these loads and identify the maximum or critical force levels that may lead to failure of the enclosure or screws.

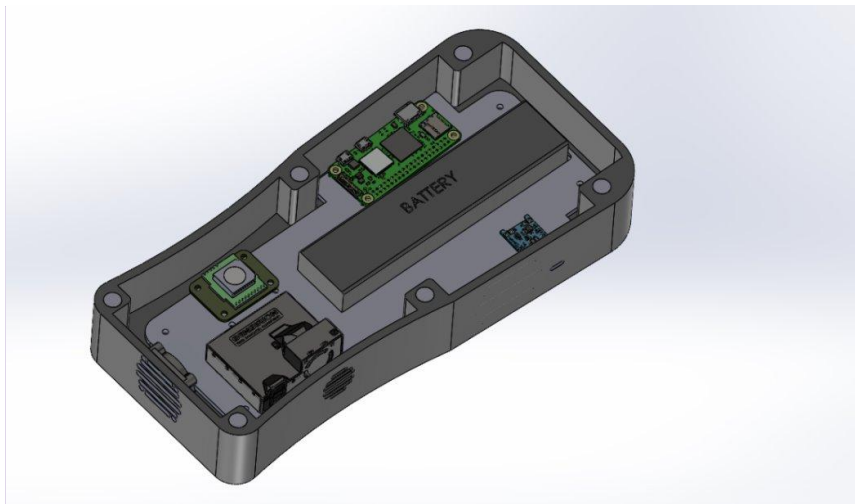
❖ Drop Testing Simulation

This method will be used to evaluate the deformation of the entire enclosure when dropped from specific heights. Drop testing will help determine the critical height at which the enclosure fails to withstand the impact, as well as identify the maximum stress values and their locations after impact.

❖ Fatigue Analysis (Recommendation)

Due to the geometric complexity of the enclosure and the likelihood of repeated loading, a fatigue life assessment may be required. This analysis will allow us to identify the stress levels associated with infinite life and the stress thresholds that correspond to a finite life cycle under repeated loading.

Modelling and Assembly



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Figure 8: Detail assembly

The model describes our initial electronic and enclosure assembly to address the issue of heat exchange between the different components in the system and ease the interchangeability of the power source or battery, we propose a second system design that minimize heat inside the system and favor a greater configuration of the electronic system and enclosure assembly.

Second system design as soon as updated

Figure 9: COMPLETE ASSEMBLY



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Challenges

- Mold creation for mass production of housing.
- Raspberry Pi Zero 2 consumes lots of power.
- Further research/testing to check sensor accuracy and precision.
- Cross-sensitivity happens to a formaldehyde sensor.

Final prototype Budget

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<u>Category</u>	<u>Cost (\$)</u>
3D printed case	
Sensirion SEN55	\$27.38
Fermion MEMS HCHO sensor	\$4.90
Raspberry Pi Zero 2W	\$18.00
Adafruit 12 mm Square Tactile Buttons	\$ 5.95
MakerHawk 3.7 V, 8000 mAh LiPo Battery	\$19.09
Soberton Inc. General Purpose Speaker	\$1.95
2.8" TFT Display Resistive Touch Screen module	\$16.39
Miscellaneous/Electronic Components	\$60.36
Hardware, i.e. Screws, Fasteners, Tools, etc.	\$94.97

Total:	\$248.99
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Recommendations for Future

For future development of this product, it is recommended that:

- Creating a mold for larger plastic components for possible mass production capabilities
- Installation of HEPA filters between sensor components to help alleviate build up and to also help reduce possible false readings for more accurate measurements
- Replace Raspberry Pi Zero 2 with a custom developed ARM processor or a PCB
- Develop a PCB board to reduce the size of the case.

Conclusion

Considering the expansion of additive manufacturing methods in several manufacturing such as automotive, aerospace and some others, it's crucial to address the issue related to the production of volatile organic compounds and formaldehyde from 3d printing processes, specifically SLA(Stereolithography) and FDM(Fused Deposition Modelling). The emissions of those different particulate compounds can be hazardous for human health when exposed to some concentrations level. The portable emission tracer is a solution suggested to inform and prevent users from an academic and individual standpoint. The project itself relies on the suction of particulate in the atmosphere to collect their percentage in the workshops in real time.

The challenge associated with its design and manufacturing were found in the management of heat transfer, dimension tolerances for portability, and the production of an adequate airflow. Using several theory from both thermodynamics and Fluid dynamics for compressible fluid we concluded that a CFM of 0.92 will be adequate for the enclosure airflow with a power as low as 0.6 W to manage the fan for cooling and suction of the outside air. Fused Deposition Modelling being, the main manufacturing process used faced

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issues with exact dimensioning and tolerances which required our teams to scale and review the design dimensions and quality of the tolerances associated.

In conclusion, the air monitoring system for detecting VOCs and formaldehyde in SLA and FDM printer environments is progressing well. Data collection and transmission via Raspberry Pi and sensors function effectively. With portability achieved and the display operational, the remaining task focuses on finalizing the enclosure for complete deployment. Further finite element analysis and drop testing will require 3d printing to make up a finished prototype to validate the optimization of the airflow and the case toughness.

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Appendix

Initial design concept

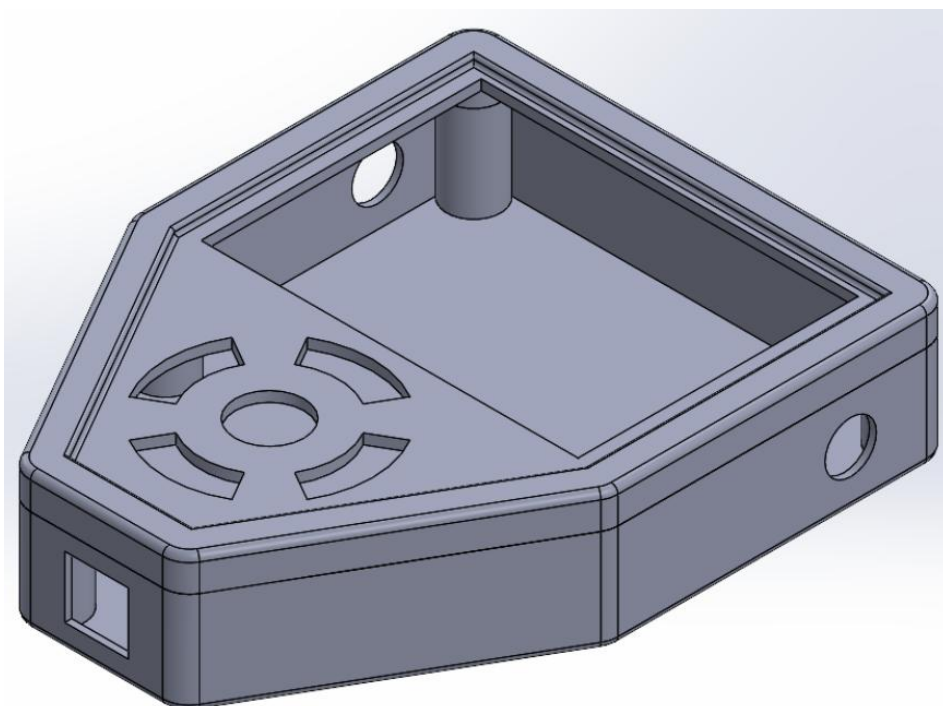


Image x: Previous design concept By: Forrest Austin

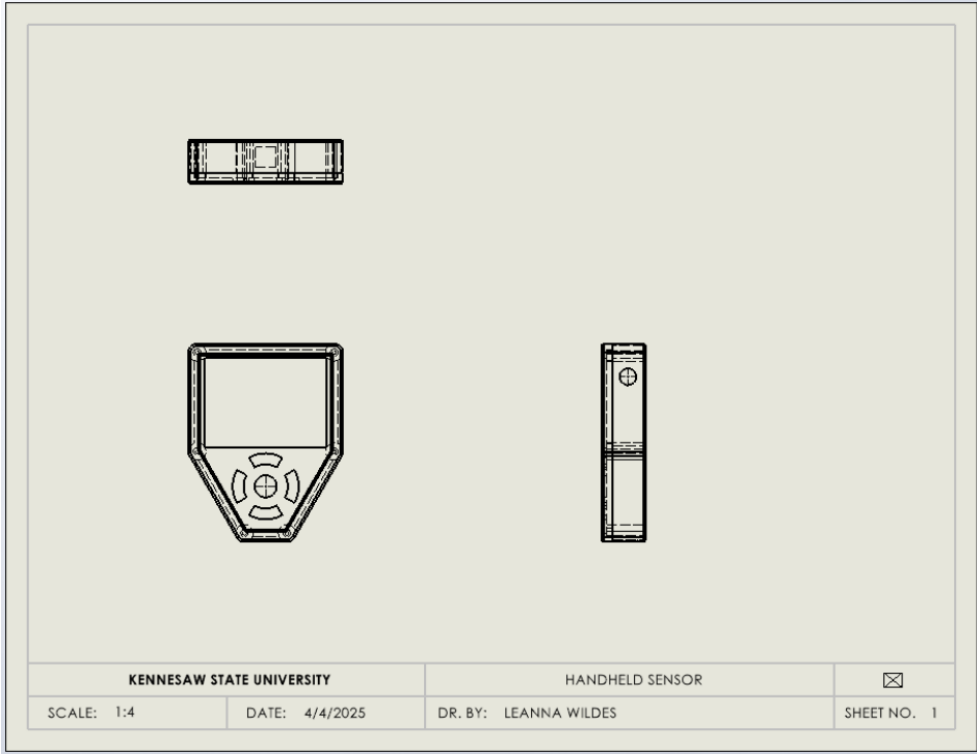


Image X: Initial design

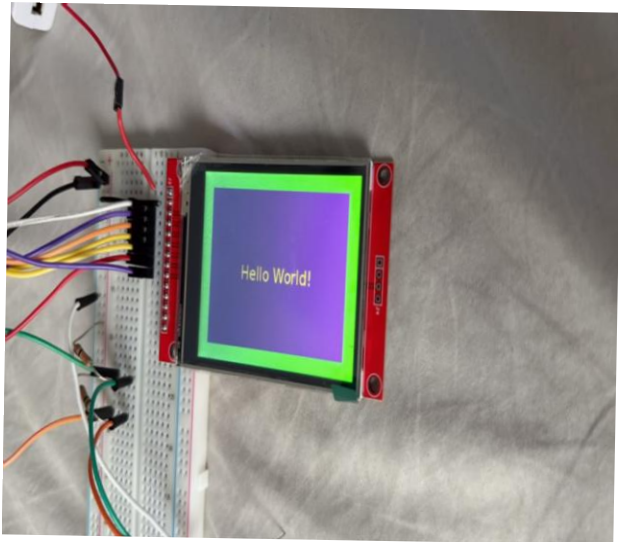


Image #: Hello World

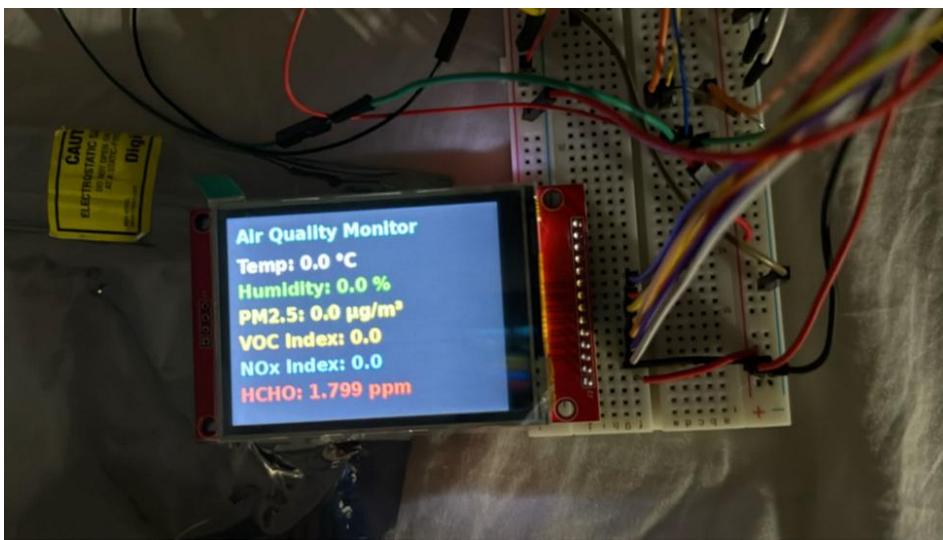
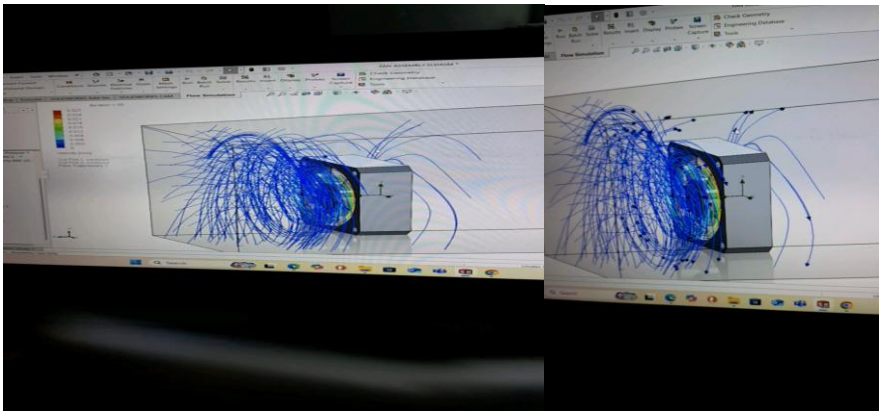


Image # :



Images & : Airflow analysis for fan

1st printed housing

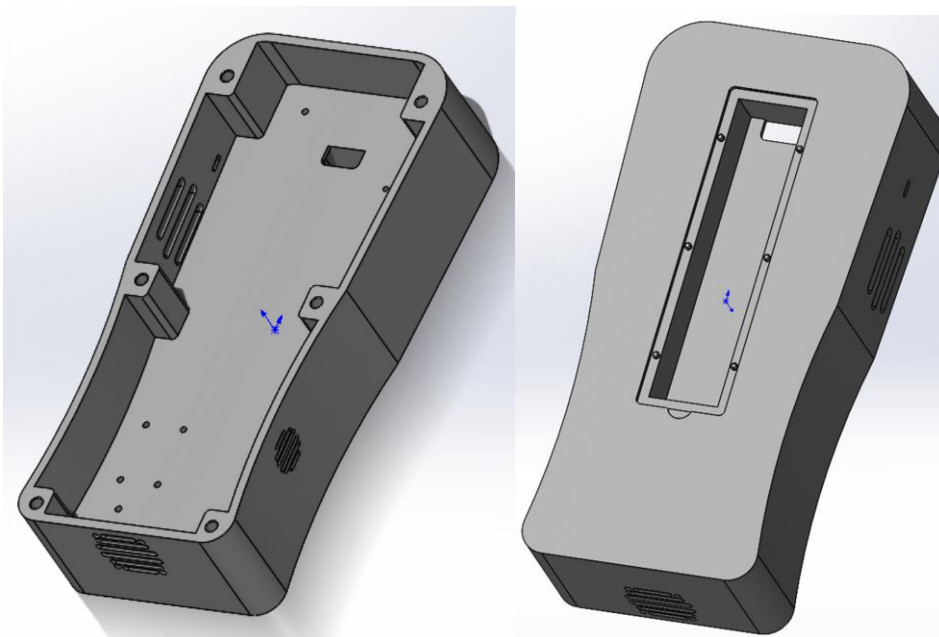
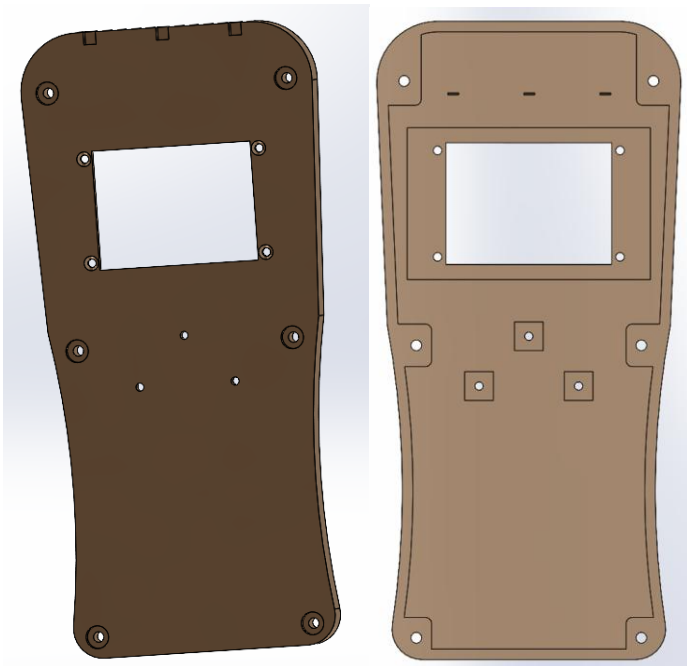


Image & : bottom of case front and back



Images & : Lid front and back

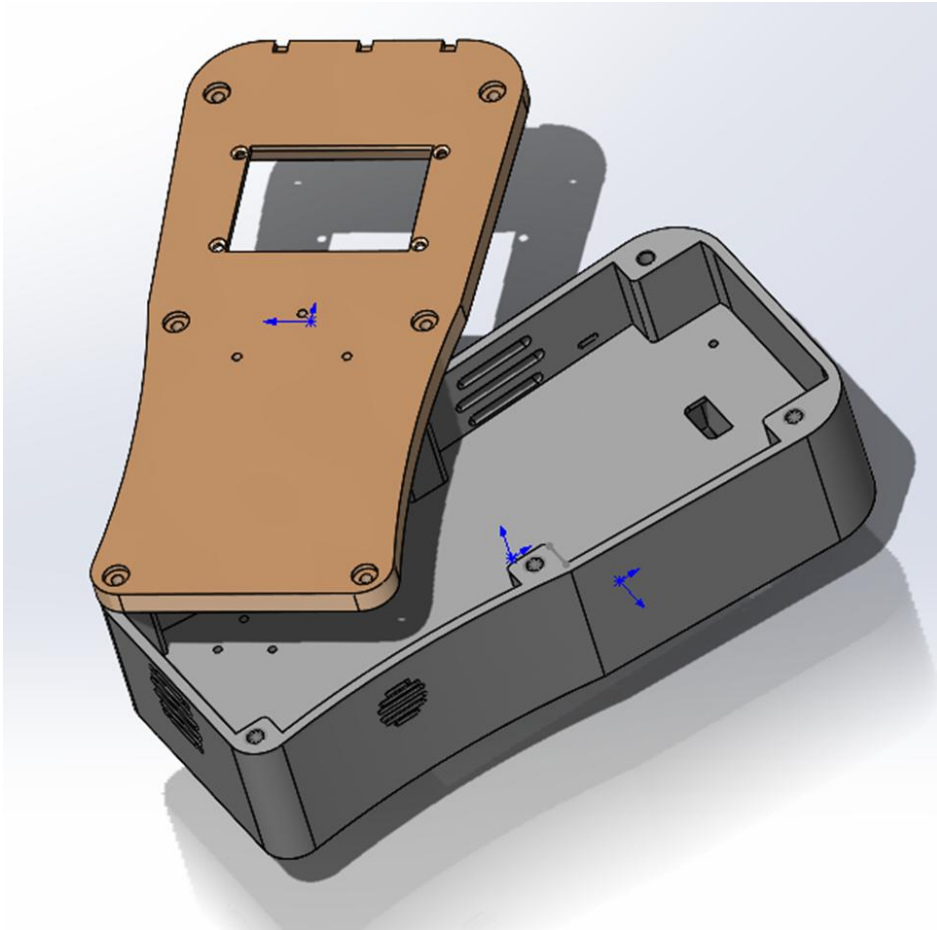


Image # :

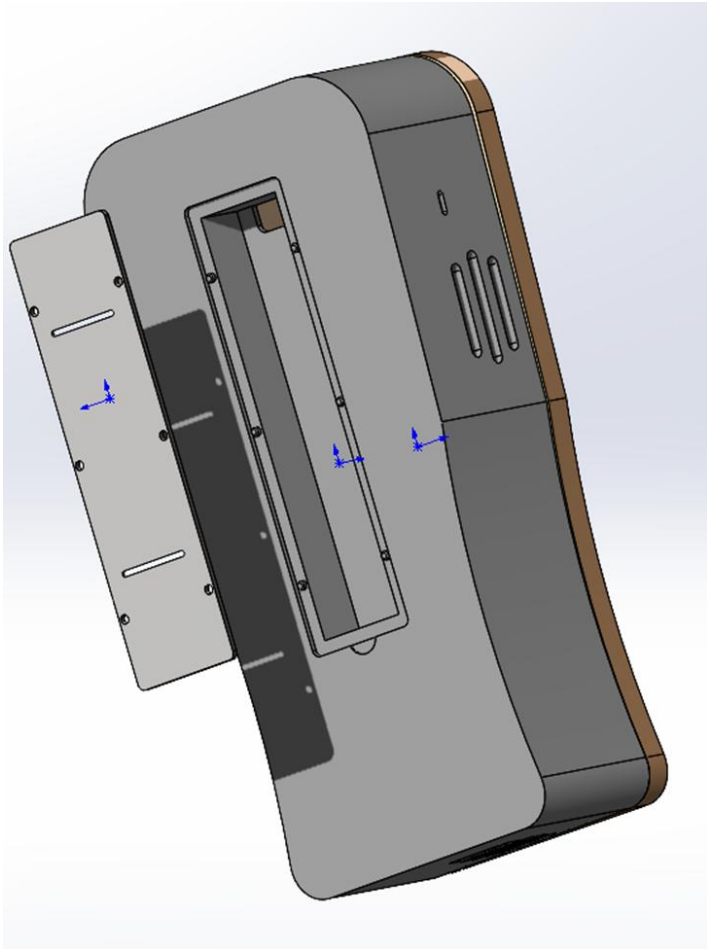


Image #: back of case and battery lid



Image #: physical print



Image #: bottom case size

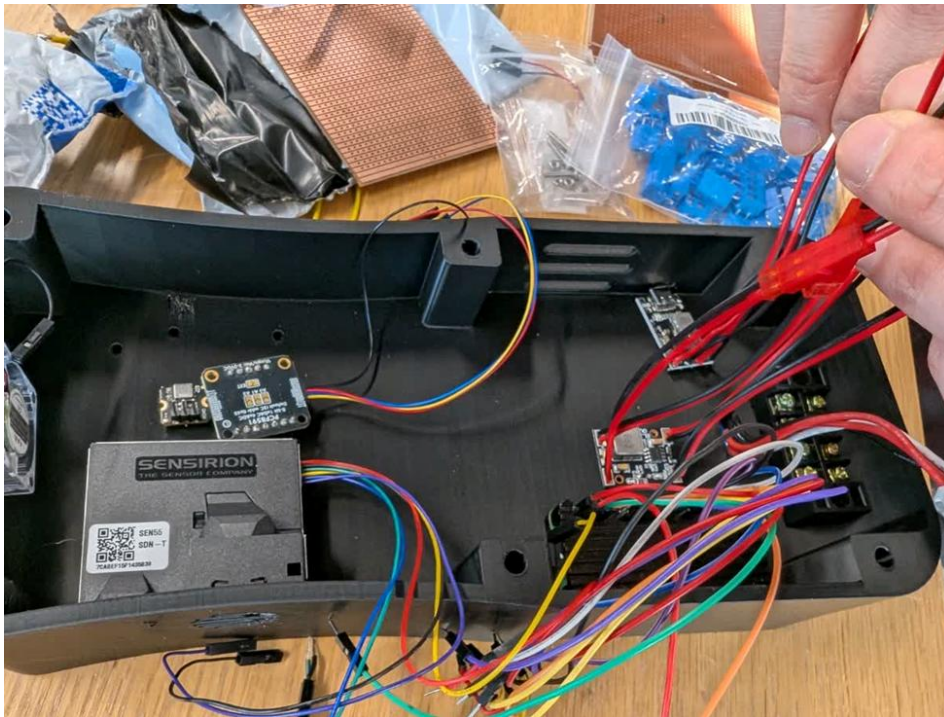


Image #: general internal layout

Soldering



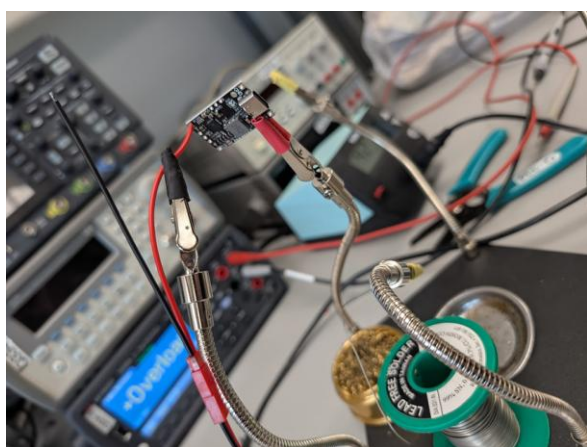
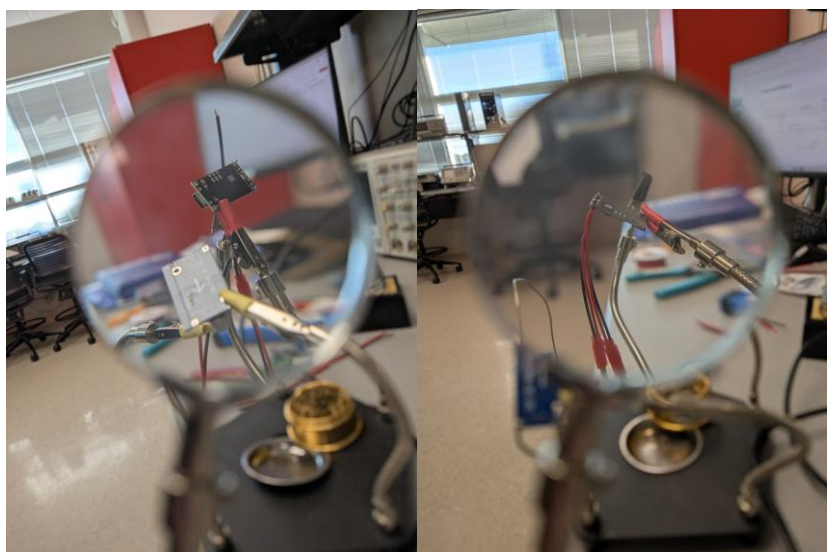
Image # : Anh Soldering



Image : soldering setup



Image # : Yvon soldering



Images & :

